

Mohammed Aziz Hossain

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EDUCATION

Umeå University

August 2023- November 2025

Master of Science in Molecular Biology

Thesis: Potential Roles of the *gdf15-gfral* Stress Signaling Pathway in Larval Zebrafish Skin.

BRAC University

June 2021- July 2023

Master of Science in Biotechnology

CGPA: 3.62/4.00

Thesis: Isolation and Characterization of *Acinetobacter baumannii* from Environmental Surface Water in Dhaka City, Bangladesh.

North South University

January 2015 - August 2018

Bachelor of Science in Microbiology

CGPA: 3.85/4.00

Thesis: Comparative Genomics of *Burkholderia pseudomallei* Strains Isolated from Bangladesh.

TECHNICAL SKILLS

Microbiology: Bacterial isolation and identification; Bacterial transformation, Bacterial growth kinetics; Virulence profile assays (biofilm formation, serum resistance); Antimicrobial susceptibility testing (MIC and MBC); Plasmid isolation; Phage purification

Molecular Biology: DNA and RNA purification; PCR; RT-PCR; qPCR; Vector modification and cloning; mRNA library preparation; RNA probe generation; In situ hybridization; CRISPR protocols; Protein extraction and purification; SDS-PAGE; Western blot; ELISA; Immunochromatography

In vitro Model: Cell culture; Tissue transfection; Drug screening; Cell viability and clonogenic assays; Flow cytometry (FACS); Immunofluorescence microscopy

In vivo Model: Zebrafish husbandry and maintenance; Generation and genotyping of mutant lines; Phenotype analysis; Whole-mount in situ hybridization

Microscopy: Fluorescence and Confocal microscopy

Data Analysis: GraphPad Prism; RStudio; SPSS; FCS Express

Bioinformatics: Rstudio; RNA Sequencing (DESeq2, Seurat); Whole genome assembly and annotation

PUBLICATIONS

1. Hossain, M.A., Fatima, N.N.E., Tushar, J.H. Isolation and characterization of *Acinetobacter baumannii* from environmental waters in Dhaka City, Bangladesh. BMC Microbiol 25, 314 (2025). <https://doi.org/10.1186/s12866-025-04029-w>

2. Mondol, S.M., Hossain, M.A. Haque, F.K.M. Comprehensive genomic insights into a highly pathogenic clone ST656 of mcr8.1 containing multidrug-resistant *Klebsiella pneumoniae* from Bangladesh. Sci Rep 15, 5909 (2025). <https://doi.org/10.1038/s41598-025-90414-4>

3. Mondol SM, Hossain, M.A, Haque FKM. Whole genome sequencing reveals the co-existence of *bla*_{PER-7}, *bla*_{ADC-52} and *bla*_{OXA-91} in multidrug resistant ST164^{pas}/ST234^{oxf} *Acinetobacter baumannii* strains in Bangladesh. Journal of Global Antimicrobial Resistance (2026). <https://doi.org/10.1016/j.jgar.2026.09.004>

4. Mondol SM, **Hossain, M.A**, Haque FKM. 2025. Draft genome analysis of a serum-resistant and fourth-generation cephalosporin (cefepime)-resistant *Acinetobacter baumannii* strain TDU5 isolated from Dhaka, Bangladesh. <https://doi.org/10.1128/mra.01332-24>

RESEARCH EXPERIENCE

Master's Thesis Student, Umeå University

January 2025- June 2025

Thesis: Potential Roles of the *gdf15-gfral* Stress Signaling Pathway in Larval Zebrafish Skin.

- Examined lipopolysaccharide (LPS)–induced activation of skin goblet cells via the *gdf15–gfral* signaling pathway in larval zebrafish.
- Quantified skin goblet cell activation by measuring phosphorylated ERK (pERK) levels as a downstream signaling readout.
- Performed qPCR to quantify *gdf15* transcript expression following LPS induced immune stimulation.
- Demonstrated that CRISPR disruption of *gfral* significantly reduced LPS-induced pERK activation, confirming pathway-specific regulation of epithelial stress responses.

Advanced Project Student, Umeå University

August 2024- January 2025

Project: Analyzing the expression profile of *fh13* genes in Zebrafish development.

- Conducted a pilot study on the temporal and spatial expression of *fh13a* and *fh13b* during zebrafish development up to 5 dpf.
- Gene expression across developmental stages using qPCR showed high expression of *fh13a* compared to *fh13b*.
- Whole-mount in situ hybridization (WISH) showed notochord-specific expression of both *fh13a* and *fh13b* genes.
- Data showed a role for *fh13a* and *fh13b* in notochord and musculoskeletal development.

Tumor Biology Course Project, Umeå University

October 2023- December 2023

Project: Assessment of ISA-2011B as a prospective drug for ER+ breast cancer.

- Overexpressed PIP5K1 α in MCF-7 breast cancer cells via plasmid transfection and assessed its effect on pAKT levels using western blots.
- Evaluated anti-proliferative effects of ISA-2011B using MTS cell viability and clonogenic assays and determined dose-dependent inhibition with an IC₅₀ of ~47–64 μ M.
- Evaluated early and late apoptosis using Annexin V/PI staining and flow cytometry, demonstrating increased late apoptosis in ISA-2011B treated cells.
- Detected DNA damage response via immunofluorescence analysis of phospho-H2AX (Ser139) following ISA-2011B treatment.

Research Assistant, BRAC University

May 2022- July 2023

Project: Characterizing the dissemination of carbapenem resistance in *Escherichia coli*, *Klebsiella pneumoniae*, and *Acinetobacter baumannii* in hospital wastewater and hospital-adjacent community water of Dhaka, Bangladesh.

- Isolated and characterized bacterial strains from hospital wastewater and adjacent community water sources.
- Performed antimicrobial susceptibility testing (MIC, disc diffusion) on ~200 isolates to determine resistance profiles.
- Conducted PCR-based detection of carbapenemase and antimicrobial resistance genes.
- Presented findings in internal research seminars and collaborated with multidisciplinary research teams.